



Solving Rational Equations

A rational equation is an equation that contains one or more rational expressions.

Methods of Solving Rational Equations

LCD method Multiply both sides of the equation by the least common denominator (LCD) of the rational expressions. Then solve the resulting equation. This method works for any rational equation.

Cross multiplication method If the equation is in a form that looks like $\frac{a}{b} = \frac{c}{d}$, you can cross multiply, then solve the resulting equation.

The method of cross multiplying is based on the following property:

$$\frac{a}{b} = \frac{c}{d} \text{ if and only if } ad = bc$$

Because rational equations have variables in denominators, you have to be careful with the domain. Any value of the variable that makes a denominator equal zero cannot be a solution, and therefore must be excluded. It's a good idea to start by identifying those values that must be excluded before you begin to solve the equation.

► **Think About It** If you forget to identify the values that must be excluded, you should discover them when you check your solutions.

EXAMPLE 1

Solve and check.

$$\frac{15}{x-6} = \frac{5}{x}$$

SOLUTION

To find the restrictions on the domain, set the denominators equal to zero and solve. The domain restrictions are $x \neq 0$ and $x \neq 6$. Now solve the equation.

$$\frac{15}{x-6} = \frac{5}{x}$$

$$15x = 5(x-6) \quad \text{Cross multiply.}$$

$$15x = 5x - 30 \quad \text{Distribute the 5.}$$

$$10x = -30 \quad \text{Subtract } 5x.$$

$$x = -3 \quad \text{Divide by 10.}$$

CHECK

$$\frac{15}{x-6} = \frac{5}{x}$$

$$\frac{15}{-3-6} \stackrel{?}{=} \frac{5}{-3} \quad \text{Substitute } -3 \text{ for } x \text{ in the original equation.}$$

$$\frac{15}{-9} \stackrel{?}{=} \frac{5}{-3}$$

$$-\frac{5}{3} = -\frac{5}{3} \checkmark \quad \text{Substituting } -3 \text{ results in a true statement.}$$

Since $x = -3$ is not in the domain restrictions, the solution set is $\{-3\}$. ■

► **Think About It** A rational equation is a proportion if it has the form

$\frac{a}{b} = \frac{c}{d}$, where $\frac{a}{b}$ and $\frac{c}{d}$ are rational expressions.

Solving Rational Equations and Checking for Extraneous Solutions

The zero product property can be used to identify domain restrictions or solve more complicated equations.

Zero Product Property

For any real numbers a and b , $ab = 0$ if and only if $a = 0$ or $b = 0$.

► **Remember** See how to apply the property with this example.

$$\begin{array}{l} (x-4)(x+1) = 0 \\ (x-4) = 0 \quad \text{or} \quad (x+1) = 0 \\ x = 4 \qquad \qquad \qquad x = -1 \end{array}$$

So the solutions to the equation are $x = 4$ and $x = -1$.

EXAMPLE 2

Solve and check.

A $\frac{3}{x^2 + 5x + 6} + \frac{x-1}{x+2} = \frac{7}{x+3}$

SOLUTION

First identify values of x that must be excluded from the domain.

$$\frac{3}{x^2 + 5x + 6} + \frac{x-1}{x+2} = \frac{7}{x+3} \quad \text{Original equation}$$

$$\frac{3}{(x+2)(x+3)} + \frac{x-1}{x+2} = \frac{7}{x+3} \quad \text{Factor the first denominator.}$$

If $x = -2$, then $x + 2 = 0$, which makes two denominators equal zero.

If $x = -3$, then $x + 3 = 0$, which makes two denominators equal zero.

So -2 and -3 must be excluded as possible solutions. Now solve the equation.

$$\frac{3}{(x+2)(x+3)} + \frac{x-1}{x+2} = \frac{7}{x+3}$$

Keep denominators in factored form to help identify the LCD.

$$\begin{aligned} (x+2)(x+3) \left[\frac{3}{(x+2)(x+3)} + \frac{x-1}{x+2} \right] \\ = (x+2)(x+3) \left(\frac{7}{x+3} \right) \end{aligned}$$

Multiply both sides by the LCD.

$$\begin{aligned} \cancel{(x+2)} \cancel{(x+3)} \frac{3}{\cancel{(x+2)} \cancel{(x+3)}} + \cancel{(x+2)} \cancel{(x+3)} \frac{x-1}{\cancel{x+2}} \\ = (x+2) \cancel{(x+3)} \frac{7}{\cancel{x+3}} \end{aligned}$$

Distribute the LCD. Divide out common factors.

$$3 + (x+3)(x-1) = (x+2) \cdot 7$$

Simplify.

$$3 + x^2 + 2x - 3 = 7x + 14$$

Multiply binomials.

Distribute 7.

$$x^2 + 2x = 7x + 14$$

Simplify.

$$x^2 - 5x - 14 = 0$$

Add $-7x - 14$ to both sides.

$$(x-7)(x+2) = 0$$

Factor the trinomial.

$$x-7=0 \quad \text{or} \quad x+2=0$$

Zero Product Property

$$x=7 \qquad \qquad x=-2$$

CHECK

Substitute 7 for x .

$$\begin{aligned}\frac{3}{x^2 + 5x + 6} + \frac{x-1}{x+2} &= \frac{7}{x+3} \\ \frac{3}{7^2 + 5 \cdot 7 + 6} + \frac{7-1}{7+2} &\stackrel{?}{=} \frac{7}{7+3} \\ \frac{3}{49 + 35 + 6} + \frac{6}{9} &\stackrel{?}{=} \frac{7}{10} \\ \frac{3}{90} + \frac{60}{90} &\stackrel{?}{=} \frac{63}{90} \\ \frac{63}{90} &= \frac{63}{90} \quad \checkmark\end{aligned}$$

So 7 is a solution.

Substitute -2 for x .

$$\begin{aligned}\frac{3}{x^2 + 5x + 6} + \frac{x-1}{x+2} &= \frac{7}{x+3} \\ \frac{3}{(-2)^2 + 5 \cdot (-2) + 6} + \frac{-2-1}{-2+2} &\stackrel{?}{=} \frac{7}{-2+3} \\ \frac{3}{0} + \frac{-3}{0} &\neq \frac{7}{1}\end{aligned}$$

Here, -2 is called an extraneous solution because it does not make the original equation true. Recall that -2 was identified as a value that must be excluded as a possible solution anyway.

The solution set is $\{7\}$.

$$\text{B } \frac{10}{n} = \frac{n+9}{n-4} - 3$$

SOLUTION

First note that 0 and 4 must be excluded as possible solutions because each value makes a denominator equal zero. Solve the equation.

$$\begin{aligned}\frac{10}{n} &= \frac{n+9}{n-4} - 3 \\ n(n-4) \frac{10}{n} &= n(n-4) \frac{n+9}{n-4} + n(n-4)(-3) \\ \cancel{n}(n-4) \frac{10}{\cancel{n}} &= n(\cancel{n-4}) \frac{n+9}{\cancel{n-4}} + n(n-4)(-3) \\ (n-4)10 &= n(n+9) - 3n(n-4) \\ 10n - 40 &= n^2 + 9n - 3n^2 + 12n \\ 2n^2 - 11n - 40 &= 0 \\ (2n+5)(n-8) &= 0\end{aligned}$$

Apply the zero product property.

$$\begin{array}{ll} 2n+5=0 & \text{or} \quad n-8=0 \\ n=-\frac{5}{2} & n=8 \end{array}$$

CHECK

Substitute $-\frac{5}{2}$ for n .

$$\begin{aligned}\frac{10}{n} &= \frac{n+9}{n-4} - 3 \\ \frac{10}{-\frac{5}{2}} &\stackrel{?}{=} \frac{-\frac{5}{2}+9}{-\frac{5}{2}-4} - 3 \\ \frac{10}{-\frac{5}{2}} &\stackrel{?}{=} \frac{\frac{13}{2}}{-\frac{13}{2}} - 3 \\ \frac{10}{1} \cdot \left(-\frac{2}{5}\right) &\stackrel{?}{=} \frac{13}{2} \cdot \left(-\frac{2}{13}\right) - 3 \\ -4 &\stackrel{?}{=} -1 - 3 \\ -4 &= -4 \checkmark\end{aligned}$$

Substitute 8 for n .

$$\frac{10}{n} = \frac{n+9}{n-4} - 3$$

$$\frac{10}{8} \stackrel{?}{=} \frac{8+9}{8-4} - 3$$

$$\frac{5}{4} \stackrel{?}{=} \frac{17}{4} - \frac{12}{4}$$

$$\frac{5}{4} = \frac{5}{4} \checkmark$$

The solution set is $\left\{-\frac{5}{2}, 8\right\}$. ■

Solving a Rate Problem

EXAMPLE 3

It takes 6 h for Pump A to fill a tank. It takes 4 h for Pump A and Pump B together to fill the same tank. How long would it take Pump B alone to fill the tank?

SOLUTION

Pump A alone can fill the tank in 6 h, so it fills $\frac{1}{6}$ of the tank each hour. Pumps A and B together can fill the tank in 4 h, so they fill $\frac{1}{4}$ of the tank each hour. Let x be the number of hours it takes Pump B to fill the tank alone. Then Pump B alone fills $\frac{1}{x}$ of the tank each hour. Now reason as follows: Pump A fills $\frac{1}{6}$ of the tank in 1 h, $\frac{2}{6}$ of the tank in 2 h, $\frac{3}{6}$ of the tank in 3 h, and so on. Pump B fills $\frac{1}{x}$ of the tank in 1 h, $\frac{2}{x}$ of the tank in 2 h, $\frac{3}{x}$ of the tank in 3 h, and so on.

Write an equation.

$$\begin{array}{ccccc}
 \boxed{\begin{array}{c} \text{the amount of the} \\ \text{tank that Pump A} \\ \text{fills in 4 h} \end{array}} & + & \boxed{\begin{array}{c} \text{the amount of the} \\ \text{tank that Pump B} \\ \text{fills in 4 h} \end{array}} & = & \boxed{\begin{array}{c} \text{one full tank} \end{array}} \\
 \downarrow & & \downarrow & & \downarrow \\
 \frac{4}{6} & + & \frac{4}{x} & = & 1
 \end{array}$$

Solve the equation.

$$\frac{4}{6} + \frac{4}{x} = 1$$

$$6x \cdot \frac{4}{6} + 6x \cdot \frac{4}{x} = 6x \cdot 1 \quad \text{Multiply both sides by the LCD.}$$

$$\cancel{6}x \cdot \frac{4}{\cancel{6}} + \cancel{6}\cancel{x} \cdot \frac{4}{x} = 6x \cdot 1 \quad \text{Divide out common factors.}$$

$$4x + 24 = 6x \quad \text{Simplify.}$$

$$24 = 2x$$

$$12 = x$$

Substitute 12 for x in the original equation to verify that 12 is the solution.

It would take 12 h for Pump B to fill the tank alone. ■

► **Think About It** For the equation $\frac{4}{6} + \frac{4}{x} = 1$, the value 0 must be excluded as a possible solution.

Solving a Work Problem

EXAMPLE 4

Mario can finish all the house chores twice as fast as his sister Ana can. Together, they can do the job in 2 h. How long would it take each of them working alone?

SOLUTION

Let x be the number of hours it takes Mario to do the chores alone. Mario does the job twice as fast as Ana, so Ana would take $2x$ hours to do the chores by herself. Then Mario does $\frac{1}{x}$ of the chores each hour, and Ana can do $\frac{1}{2x}$ of the chores each hour. Mario and Ana together can finish the chores in 2 h, so together they can do $\frac{1}{2}$ of the job each hour. Write an equation that combines Mario's and Ana's labor for each hour.

$$\frac{1}{x} + \frac{1}{2x} = \frac{1}{2}$$

Solve the equation.

$$\frac{1}{x} + \frac{1}{2x} = \frac{1}{2}$$

$$2x \cdot \frac{1}{x} + 2x \cdot \frac{1}{2x} = 2x \cdot \frac{1}{2} \quad \text{Multiply both sides by the LCD.}$$

$$2x \cdot \frac{1}{x} + 2x \cdot \frac{1}{2x} = 2x \cdot \frac{1}{2} \quad \text{Divide out common factors.}$$

$$2 + 1 = x \quad \text{Simplify.}$$

$$3 = x$$

Substitute 3 for x in the original equation to verify that 3 is the solution. It would take 3 h for Mario to do all the house chores alone, and it would take twice as long, or 6 h, for Ana to do the job alone. ■

Solving for a Variable in a Real-World Rational Equation

Solving for a variable in a real-world rational equation does not always produce a numerical value. However, the steps used to solve the equation are the same.

EXAMPLE 5

One way to express the work formula is in the form $\frac{t}{a} + \frac{t}{b} = 1$, where t is the time it takes two people to complete the work, a is the time it takes Person A to complete the work alone, and b is the time it takes Person B to complete the work alone. Solve this equation for t .

SOLUTION

Multiply both sides of the equation by the least common denominator ab , making it easier to solve for t .

$$\frac{t}{a} + \frac{t}{b} = 1$$

$$ab \cdot \left(\frac{t}{a}\right) + ab \cdot \left(\frac{t}{b}\right) = ab \cdot 1 \quad \text{Multiply both sides by the LCD.}$$

$$ab\left(\frac{t}{a}\right) + ab\left(\frac{t}{b}\right) = ab \quad \text{Divide out common factors.}$$

$$bt + at = ab \quad \text{Simplify.}$$

$$t(b + a) = ab \quad \text{Distributive Property}$$

$$t(a + b) = ab \quad \text{Commutative Property}$$

$$t = \frac{ab}{(a + b)} \quad \text{Divide both sides by } (a + b).$$

When solved for t , the expression $\frac{t}{a} + \frac{t}{b} = 1$ is $t = \frac{ab}{(a + b)}$. ■